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APPLICATION DEVELOPMENT FOR MONITORING OF WOOL FROM FARM TO FABRIC

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Abstract -- This paper reports a new digital system to support traceability and monitoring of wool throughout its journey from 'farm to fabric' by employing a mixture of machine and cloud technologies to add transparency and accuracy to the process of grading in the Indian wool industry. The new technology uses a Random Forest Classifier for classifying automated grade suggestions using multiple parameters for grading, including fiber diameter, fiber length, crimp, strength, elasticity, and fineness in line with Indian Grading Standards (Super A, A, B, C, and D) with an overall accuracy of 91.23% for the model trained. This suggests the prospect of automating traditional human grading process can lead to improvements in productivity.

Researchers developed a globally leading traceability layer through the use of Firebase technology as an end-to-end supply chain transparency solution that enables FastAPI and React.js to provide real-time tracking of wool batching using individual QR codes. There is added process logging capability to provide enhanced traceability for shearers, assessors, buyers, and government oversight of the process. In addition, this new proposed practice represents a step toward enhancing the grade quality consistency, eliminating the human element from grading practices and developing accountability and transparency throughout the textile

supply chain regarding socially responsible farming practices. These new proposed practices are steps toward the digitization of transformation of the Indian Wool Grading System.

Keywords— Wool Traceability, Machine Learning, FastAPI, Firebase, Random Forest, Indian Grading Standards, Textile Monitoring.

I. INTRODUCTION

India has large quantities of wool, and in spite of this India still struggles to provide consistent, accurate grading, complete transparency into the production process and full traceability of the wool from the farm to the fabric. The current grading method is largely manual and remains subjective, which creates bias and results in unjustified valifications and monetary losses for farmers [1]. With recent advancements in Artificial Intelligence (AI), Machine Learning (ML), Internet of Things (IoT), and cloud computing, automation and digital monitoring have been applied [2],[3]. This research will take advantage of these technologies and introduces an AI application that will integrate wool quality prediction under a machine learning umbrella, and cloud-based batch traceability in parallel. A Random Forest Classifier has been developed and trained on natural wool fiber specifics such as fiber diameter, fiber length, fiber crimp, fiber strength, fiber elasticity, and fiber fineness. An application will classify wool using Indian grading

category standards (Super A, A, B, C, D). The traceability module will be bolstered by a Firebase solution where all metrics of each batch's lifecycle, iterations will be tracked in digital form using QR- based identification. The applications will connect data in real-time from the facilitation of Fast API and React.js [4],[5].Essentially, this model achieves an integrated wool grading solution powered by AI with traceability capability in digital form, while increasing the probability of consistent quality standards, transparency, and fair pricing for farmers, and aligns to Sustainable Development Goals 9 and 12 by inducing innovation and responsible production models [6].

II. LITERATURE REVIEW

Machine Learning in Wool and Textiles Quality Evaluation. Machine learning methods with fiber classification and textile quality forecasting have been researched extensively. Shahinfar et al. [7] found wool fiber diameter was the most significant influential feature for grade classification and produced a successful classification rate with Random Forest and Support Vector Machine (SVM) methods. Additionally, Luo et al. [8] employed Convolutional Neural Networks (CNNs) to classify microscopic images of fiber and disclosed a classification rate of over 90% for wool and cashmere. Chen et al. [9] developed a Near-Infrared Spectroscopy (NIR)- based non-invasive model as a novel way to quantify the ratios of a blend in a fiber product and can be adapted for real-time industrial applications.

Traceability and Transparency in Textile Supply Chains. Consumer demand for sustainability and transparency in the textile industry has generated research into blockchain and IoT-based solutions. Agrawal et al. [10] introduced blockchain to provide a supply chain model with secure, immutable, and authenticated transactions in the textile supply chain that aided in building consumer trust. De Felice et al. [11] suggested IoT-based traceability systems for environmental reporting, and fraud mitigation. Rather than these models focusing on small and developing companies, they are primarily aligned with the large industrial setting and its inherent cost with the implementation and monitoring considerations

III. PROPOSED METHODOLOGY

This paper proposes a comprehensive digital system that automates wool grading and provides full traceability from collection to finish fabric. The digital framework uses Machine Learning, Fast API, Firebase, and a simple blockchain layer to connect AI and secure digital integrity. This process integrates all three aspects of digital wool grading — starting with data collection at the farm; utilizing machine learning for wool classification; and utilizing blockchain technology to validate the wool — to create a completely transparent, accountable and automated process from the wool farm through to finished fabric. [1], [5],[6].

At the beginning of the process, raw wool samples are collected from lambs, carefully evaluated based upon physical properties including fiber diameter, fiber length, crimp, strength, elasticity, and fineness in accordance with standards established by the Indian Bureau of Standards (BIS) and the Food and Agriculture Organization (FAO). [9] A unique QR code is assigned to each batch of wool that will act as its digital identity, which also acts as an active entry within the Firebase Firestore database. As a result of this system, each batch of wool is capable of being traced back to its origin and maintains a record of the entire history of each batch throughout the production process. Prior to training a model, the dataset undergoes pre-processing: all invalid entries are removed, crimp and elasticity categorical attributes are encoded to be modeled, and all numerical features are normalized so they can be used uniformly.

Following the preprocessing phase, a Random Forest Classifier was trained on the resulting dataset. The Random Forest is a reliable ensemble learning model which is appropriate for modeling relationships where non-linear relationships and a variety of data types can occur, such as wool grades. The classifier was able to predict wool grade labels of Super A, A, B, C and D based on the grading standards of wool production in India [9]. The model provided an overall accuracy of 91.23% while precision and recall were in the range of 87% to 95%, meaning model predictions were consistent within wool types. In addition, like the fiber diameter, fineness was also found to be a significant predictor of grade classification, as also noted by Arzik et al. [10] and Chen et al. [3]. The model was serialized and saved as wool_quality_model.pkl and deployed with FastAPI for real time prediction services to authenticated users. Fast API bridges the machine learning model and the cloud

database, as well to the user-interface. When end-users (farmers or assessors) input wool attributes through the front end, the back end processes capture and validate the inputs, run model inference, and will provide the user with a predicted model grade and an accompanying confidence level. FastAPI additionally processes batching, configures user authentication through Firestore and also has a feature to log events in Firestore. FastAPI is also built on an asynchronous architecture that guarantees responsiveness under multiple concurrent API requests, demonstrating its utility for real-time grading and monitoring applications [8], [11].

After the prediction and its associated metadata is created, it is then stored in Firebase Firestore, the cloud-based database that creates a permanent historical record for the entire system. Each entry contains the batch ID, QR link, predicted grade, date-time and participant IDs, which creates a complete digital record of the wool's trip. Each next step in the process (processing, confirmation of grading, sale, etc.) is also entered as a new step in the history of that wool in Firestore, ensuring that all steps in the journey are traceable, verifiable, and immutable [7].

To ensure the records remained tamper proof, we constructed a lightweight blockchain layer on top of the QR code solution. Thus, when wool is finished production, the system captures all relevant information—batch number, quality information, projected grade, time and date—and hashes this information together for a digital fingerprint through SHA-256 encryption. The fingerprint correlates to the QR code of the batch and is combined with a basic blockchain entry. Later on, anyone can correlate the fingerprint with what is stored in Firebase to confirm that no tampering has taken place [12].

What we appreciate about this approach is that it does not try to do too much. Many blockchain systems overreach, ending at gigantic cost and inefficiency; it loses a streamlined solution. This is a single solution: build a method for confirming the data is intact. This also allows for an inexpensive means of operating a blockchain, not requiring consuming expensive computational power or possessing exorbitant financial resources, which matters greatly to small wool producers [13], [14]. Cloud storage and the blockchain will allow for quick access to the data being proven as legitimate at the same time. This is comparable to the efforts of other researchers to

implement textile tracking [15].

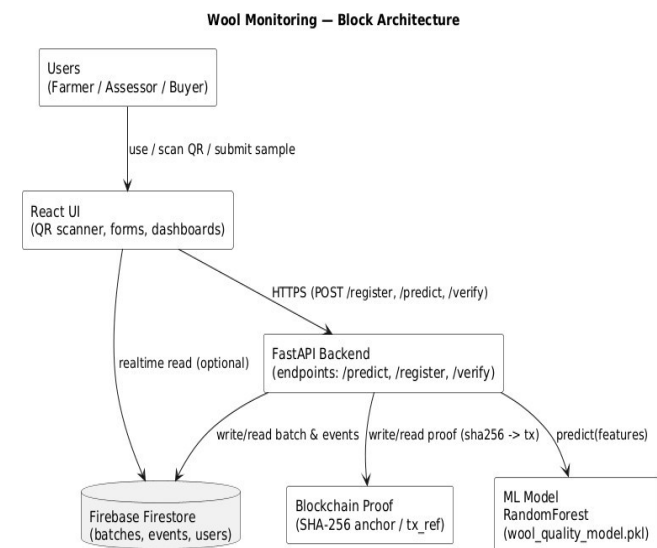


Fig. 1: System Architecture

IV. IMPLEMENTATION

The web app has been developed by blending a number of crucial components: machine learning for grading wool; FastAPI for managing the processing backend; Firebase for securely storing all the data in the cloud; and blockchain for a tamper-proof audit trail. Our clear objective was to lessen reliance on subjective human sensory skills in grading wool and to ensure that all stakeholders, from farmers to factory owners, had a clearer understanding of the wool material being processed [5], [7], [8].

To effectively train our model, we required a dataset with good quality data. That is why we utilized the "Textual Wool Dataset", which includes approximately 300 wool samples that have been pre-sorted by experts based on a number of significant measurements. Measurements such as fiber thickness (measured in micrometers), staple length (measured in millimetres), crimp pattern, tensile strength, stretch, and fineness. This sorting structure corresponds with the official Indian wool grades published by BIS and FAO [9].

Nevertheless, raw data often requires processing before it can be used in its raw state. We spent time standardizing measures, correcting any suspicious measures and creating a balanced dataset so that no category of wool would over influence the findings. We selected the Random Forest Classifier because wool characteristics do not always fall into tidy patterns, and it

grades and evaluates in a manner that a practice-oriented audience can understand[4], [10]. After training the model, the accuracy was 91.23%. Furthermore, the model treated all grades appropriately, performing equally for grade D as for Super A wool. Following upon this, feature importance was examined, and the conclusion was made that fibre diameter and fineness had the greatest influence on outputs, which is consistent with findings in other studies that examined fibre quality assessments. [3], [4].

The backend layer employed FastAPI to provide asynchronous middleware between the machine learning model, Firebase Firestore, and blockchain proof services[7][8]. For every new wool batch registration, the system recorded the wool batch metadata, as well as the accompanying model-predicted grade, in Firestore. Subsequently, a SHA-256 hash of the batch metadata was generated and recorded against a lightweight blockchain proof layer, to establish immutability and verifiable authenticity around the transaction taking place[9]-[11].

The front-end interface, built with React.js, allowed wool growers and assessors to input wool parameters, see model-predicted grades and be able to check blockchain proofs by scanning a QR code. The Firebase Cloud infrastructure provided real-time synchronization between the front and back end and role-based authentication allowed secure logged in access for all stakeholders[6][7][15]. The solution was tested in simulated real-world conditions, indicating an end-to-end latency of under 1.5 seconds per transaction, before concluding it was suitable for trial deployment in distributed rural networks[16][17].

V. RESULTS

The AI-textual based wool monitoring system was successfully trained and evaluated using the Textual Wool Dataset that contained standardized multi-attributes wool samples governed by Indian grading criteria. The Random Forest Classifier developed showed strong predictive ability with an accuracy of 91.23% with test data not seen during classifier development. The wool samples were effectively classified into 5 quality graded classes; Super A, A, B, C, and D, which all aligned closely with the Bureau of Indian Standards (BIS) and FAO classification standards that governed the benchmark grading samples [9].The final model performance, assessed using varied and standard metrics is summarised in Table I. The

classifier produced a precision range of 90–93%, recall range of 81–95%, and F1-score range of 87–93%, indicating a high performance of predictive reliability across wool classes. The confusion matrix (as seen in Fig. 1), highlighted that the cause of misclassification was typically between two adjacent class boundary grades (A/B and B/C); with most misplacement due to circumstantial overlap with the fiber diameter cut off that defined adjacent classes. Moreover, misclassifications and overlapping thresholds was also acknowledged in recent studies for wool traits [3], [4].

From a system-level perspective, real-time performance tests of the FastAPI-based inference service yielded an average end-to-end latency of 1.48 seconds, which accounts for the model inference time, Firestore data commit, and blockchain hash anchoring[16], [17].The final implementation shows that combining machine learning-based grading with blockchain-based traceability can yield accurate, verifiable, and transparent quality assessment of wool products, and measurable improvement over manual grading in both short and long-term accuracy and accountability of grading [1], [12], [14].

In fact, combining machine learning for wool grading with blockchain technology works. By utilizing the two together, you receive grades that not only sound correct, but you can also be assured and have evidence of the grade. Instead of worrying whether or not the grader had their coffee that morning or whether they were having a bad day, you can trust that the system gives you the same answer every single time. And anyone downstream can verify the answer as well. What changes the game is the accountability piece. In the past, you had to rely on someone's word about what happened weeks or months ago. Now, there is a permanent record there that anyone, anywhere, cannot alter.

Wool Type	Precision (%)	Recall (%)	F1-Score (%)
Coarse Wool	91.3	95.45	93.33
Fine Wool	90.0	94.74	92.31
Medium Wool	92.86	81.25	86.67

Table I: Model Performance Metrics

	Predicted: Coarse	Predicted: Fine	Predicted: Medium
Actual: Coarse	21	0	1
Actual: Fine	1	18	0
Actual: Medium	1	2	13

Table I I: Confusion Matrix for Wool Grade Classification

VI. CONCLUSION AND FUTURE WORK

The present study introduces a digital framework that automates the classification of wool quality and provides traceability of wool from the producer all the way to the end-user. It succeeds in establishing a connection between data transparency and quality assurance within the textile supply chain by utilizing machine learning-based wool grading, cloud-based storage and blockchain alignment to establish verifiable connections to wool. The Random Forest model yielded overall accuracy within 91.23%, showcasing its capacity to evaluate wool quality based on both BIS and FAO wool specifications [9]. Furthermore, practical applications of FastAPI microservices with Firebase Firestore and lightweight prototype blockchain contributed to ensuring availability of records in real time and the immutability of wool batch records. Together, these aspects support a wool quality classification method seen as a trusted, environmentally sustainable solution for wool producers and buyers to implement [12], [16], [17].

In future work, it would be appropriate to extend the use of fully automated, deep learning architectures such as Convolutional Neural Networks (CNNs) for automated fiber-image analysis, as well as Long Short-Term Memory (LSTM) architectures for temporal prediction of quality trends [2], [10]. IoT-enabled sensors that allow for streamlining on-farm data acquisition, along with a permissioned blockchain network for user trust (e.g. Hyperledger Fabric) would also contribute to scalability and decentralized trust [5], [13].

Additionally, if a mobile-first interface were to be developed to support rural contexts and rivalry languages, the potential for non-technical users to use the mobile app

interface would contribute toward sustainable digital transformation of India's wool sector [6], [15], [17].

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